UMSL Mechanical Engineering Courses by ME Subdiscipline

School of Engineering, 2025-2026 Catalog

Fluid Dynamics and Heat Transfer (18 credits)

Course	Title	Cr	Description
ENGR 2330	Introduction to Thermodynamics	3	Properties of substances, first and second laws, entropy, ideal gas — foundation for all thermal/fluid courses
ENGR 3300	Applied Thermodynamics	3	Power/refrigeration cycles, gas mixtures, psychrometrics, combustion — builds on ENGR 2330
MENG 3370/L	Fluid Mechanics + Lab	4	Fluid statics, Bernoulli, conservation laws, pipe flow, external flow, dimensional analysis; lab experiments
MENG 3380/L	Heat Transfer + Lab	4	Conduction (steady/transient), convection (forced/natural), radiation, heat exchangers; lab experiments
MENG 4450	Computational Fluid Dynamics	3	Numerical solution of governing equations, discretization, turbulence modeling, commercial CFD software
MENG 4460	Renewable Energy Systems Lab	1	Solar, wind, and fuel cell energy systems — applied thermal/fluid concepts in renewable energy context

Solid Mechanics and Materials (16 credits)

Course	Title	Cr	Description
ENGR 2310	Statics	3	Equilibrium, trusses/frames, centroids, friction, moments of inertia — prerequisite for all solid mechanics
ENGR 2332	Mechanics of Materials	3	Stress/strain, axial loading, torsion, bending, shear, combined loading, beam deflection, buckling
MENG 3340	Properties of Material and Testing	3	Crystal structure, phase diagrams, mechanical properties, failure modes, materials selection, heat treatment
MENG 3360	Machine Design and Manufacturing	3	Machine element design (shafts, bearings, gears), fatigue analysis, safety factors, DFM
MENG 4440	Intro to Composite Materials	3	Fiber-reinforced composites: micromechanics, laminate theory (CLT), failure criteria, manufacturing
MENG 3390	Product Dev. and Prototyping Lab	1	Product development, additive manufacturing, reverse engineering — applied solid mechanics in prototyping

Manufacturing and Design (13 credits)

Course	Title	Cr	Description
MENG 1204	ME 3D Design (CAD)	2	CAD fundamentals using SolidWorks: sketching, solid modeling, assemblies, engineering drawings, 3D printing
MENG 3360	Machine Design and Manufacturing	3	Machine element design combined with manufacturing processes overview, design for manufacturing
MENG 3390	Product Dev. and Prototyping Lab	1	Product development lifecycle, additive manufacturing, reverse engineering, rapid prototyping
MENG 4490	Advanced Manufacturing	3	CNC machining, advanced additive mfg, automation/robotics, Industry 4.0, smart manufacturing, quality control
MENG 4980	Senior Design I	2	Capstone design: project planning, concept generation, preliminary analysis, team-based open-ended design
MENG 4990	Senior Design II	2	Capstone continued: detailed analysis, fabrication, testing, validation, technical documentation

Controls, Dynamics, and Instrumentation (13 credits)

Course	Title	Cr	Description
ENGR 2320	Dynamics	3	Particle/rigid body kinematics and kinetics, work-energy, impulse-momentum — foundation for controls
EENG 2310	Circuit Analysis I	3	DC/AC circuits, Kirchhoff's laws, transients, phasors — provides electrical foundation for instrumentation
MENG 3330	Instrumentation and Measurement	3	Sensors, signal conditioning, data acquisition, measurement uncertainty — bridge between circuits and ME
MENG 3350/L	System Dynamics and Control + Lab	4	System modeling, Laplace transforms, PID control, frequency response, root locus; controller design lab

Mathematics Foundation (21 credits)

Course	Title	Cr	Description
MATH 1800	Analytic Geometry and Calculus I	5	Limits, derivatives, integrals of single-variable functions; applications to rates of change, optimization, area
MATH 1900	Analytic Geometry and Calculus II	5	Techniques of integration, sequences, series, parametric equations, polar coordinates
MATH 2000	Analytic Geometry and Calculus III	5	Vectors, partial derivatives, multiple integrals, vector calculus (Green's, Stokes', Divergence theorems)
MATH 2020	Introduction to Differential Equations	3	First/second order ODEs, Laplace transforms, systems of ODEs — essential for fluids, controls, heat transfer
MATH 1320	Intro to Probability and Statistics	3	Descriptive statistics, probability distributions, hypothesis testing, regression, confidence intervals

Physics and Chemistry (14 credits)

Course	Title	Cr	Description
PHYS 2111/L	Physics I: Mechanics and Heat + Lab	5	Calculus-based mechanics: kinematics, Newton's laws, work-energy, momentum, rotational dynamics, oscillations
PHYS 2112/L	Physics II: E&M;, Optics + Lab	5	Calculus-based E&M;: electric fields, Gauss's law, circuits, magnetism, Faraday's law, EM waves, optics
CHEM XXXX	General Chemistry for Engineering	4	Atomic structure, bonding, stoichiometry, thermochemistry, states of matter, solutions, equilibria

General Education and Other Requirements (25 credits)

Course	Title	Cr	Description
ENGL 1100	First-Year Writing	3	College-level writing skills: argumentation, research, rhetoric, academic discourse
ENGL 3130	Technical Writing	3	Writing for engineering audiences: reports, proposals, documentation, technical communication
PHIL 2259	Engineering Ethics	3	Ethical theories, engineering codes of ethics, case studies, AI ethics, professional responsibility
ENGR 1414	Elementary Engineering Design	2	Introduction to design process, project planning, teamwork, systems integration — first-year experience

Course	Title	Cr	Description
ENGR 2022	Engineering Economics and PM	3	Time value of money, economic analysis, depreciation, project management fundamentals
ENGR 4400	FE Exam Review	1	Comprehensive review of all ME topics for the Fundamentals of Engineering licensing exam
INTDSC 1003	First Year Experience	1	University transition: faculty expectations, support services, student life
	Core — American History/Government	3	General education requirement in American history or government
	Explore — Social Sciences (x2)	6	Two social science electives, one with global perspectives component
	Explore — Humanities & Fine Arts (x2)	6	Two humanities or fine arts electives for breadth
	Core — Communication Proficiency	3	Oral or written communication proficiency requirement

Note: Some courses appear in multiple categories (e.g., MENG 3360 appears in both Solid Mechanics and Manufacturing) because they span subdisciplines. This is consistent with the DEEP project's finding that traditional subdiscipline boundaries are somewhat artificial. The four ME subdiscipline tables above contain all ME-specific and engineering core courses. The remaining tables list the math/science foundation and general education courses that complete the 125-credit BSME degree.